

C++ LAB PRACTICAL MANUAL

Practical 1: Write a C++ program to print "Hello World"

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Hello World";
    return 0;
}
```

Practical 2: Write a C++ program to implement basic arithmetic operations

```
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;
    int sum, diff, prod, mod;
    float div;

    cout << "Enter number 1: ";
    cin >> num1;

    cout << "Enter number 2: ";
    cin >> num2;

    sum = num1 + num2;
    diff = num1 - num2;
    prod = num1 * num2;
    div = (float)num1 / num2;
    mod = num1 % num2;

    cout << "Sum = " << sum << endl;
    cout << "Difference = " << diff << endl;
    cout << "Product = " << prod << endl;
    cout << "Division = " << div << endl;
    cout << "Modulus = " << mod << endl;

    return 0;
}
```

Practical 3: Write a C++ program to check whether a number is even or odd

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter the number to be checked: ";
    cin >> num;

    if (num % 2 == 0)
        cout << "Number is Even";
    else
        cout << "Number is Odd";

    return 0;
}
```

Practical 4: Write a C++ program to check whether a number is positive, negative or zero

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter the number to be checked: ";
    cin >> num;

    if (num > 0)
        cout << "Number is Positive";
    else if (num < 0)
        cout << "Number is Negative";
    else
        cout << "Number is Zero";

    return 0;
}
```

Practical 5: Write a C++ program to find the greatest among three numbers

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;

    cout << "Enter first number: ";
    cin >> a;

    cout << "Enter second number: ";
    cin >> b;

    cout << "Enter third number: ";
    cin >> c;

    if (a >= b && a >= c)
        cout << "Greatest number is " << a;
    else if (b >= a && b >= c)
        cout << "Greatest number is " << b;
    else
        cout << "Greatest number is " << c;

    return 0;
}
```

Practical 6: Write a C++ program to check whether a character is a vowel or consonant using switch case

```
#include <iostream>
using namespace std;

int main()
{
    char ch;

    cout << "Enter the character to be checked: ";
    cin >> ch;

    switch (ch)
    {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << "Character is a Vowel";
            break;

        default:
            cout << "Character is a Consonant";
    }

    return 0;
}
```